



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: BCA
Semester: II
Name of the Paper: Fundamentals of Python Programming
Paper Code: TBC211
Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Define the properties of a string in Python. Demonstrate any SIX predefined string functions with examples. (CO2)

OR

b. What are **lambda functions**? Explain their syntax, advantages, and applications with suitable examples. (CO2)

Q2. (10 Marks)

a. Write a Python program that takes marks of 5 subjects, calculates the percentage, and prints the grade using if-elif-else. (CO1)

OR

b. Write a program:

1. Calculate the factorial of a given number using while loop.
2. Print all the prime numbers from the given list. (CO1)

Q3. (10 Marks)

a. Explain the built-in data types in Python. Support your answer with relevant examples. (CO1)

OR

b. What is order of operations in Python? Explain operator precedence and associativity with suitable examples. (CO1)

Q4. (10 Marks)

a. Write a recursive function to compute the sum of digits of a given number.

Example: `sum_digits(1234) → 10`. (CO2)

OR

b. What is shorthand if statement? Explain its syntax and advantages with examples. Compare it with traditional conditional statements. (CO1)

Q5. (10 Marks)

a. Explain different types of arguments used in Python functions. Elaborate your answer with suitable example. (CO2)

OR

b. Explain why strings are immutable in Python. Discuss how immutability affects string operations with examples. (CO2)